```
1  % planet(Name, PositionFromSun, Type, NumberOfMoons)
2  planet(mercury, 1, rocky, 0).
3  planet(venus,   2, rocky, 0).
4  planet(earth,   3, rocky, 1).
5  planet(mars,    4, rocky, 2).
6  planet(jupiter, 5, gas_giant, 95).
7  planet(saturn,  6, gas_giant, 146).
8  planet(uranus,  7, ice_giant, 28).
9  planet(neptune, 8, ice_giant, 16).
10
11 gas_giant(P) :- planet(P, _, gas_giant, _).
12 has_moons(P) :- planet(P, _, _, M), M > 0.
13 closer_than(P1, P2) :- planet(P1, A, _, _), planet(P2, B, _, _), A < B.
14 rocky_planets(L) :- findall(P, planet(P, _, rocky, _), L).
15
16
17
```

```
?- gas_giant(P).
```

P = jupiter
P = saturn

```
% planet(Name, PositionFromSun, Type, NumberOfMoons)
planet(mercury, 1, rocky, 0).
planet(venus,   2, rocky, 0).
planet(earth,   3, rocky, 1).
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rocky_planets(L) :- findall(P, planet(P, _, rocky, _), L).


```

```
?- has_moons(P).
```

P = earth
P = mars
P = jupiter
P = saturn
P = uranus
P = neptune

```
% planet(Name, PositionFromSun, Type, NumberOfMoons)
planet(mercury, 1, rocky, 0).
planet(venus,   2, rocky, 0).
planet(earth,   3, rocky, 1).
planet(mars,    4, rocky, 2).
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gas_giant(P) :- planet(P, _, gas_giant, _).
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rocky_planets(L) :- findall(P, planet(P, _, rocky, _), L).
```

```
?- closer_than(earth, mars).
```

true

1

```
% planet(Name, PositionFromSun, Type, NumberOfMoons)
planet(mercury, 1, rocky, 0).
planet(venus,   2, rocky, 0).
planet(earth,   3, rocky, 1).
planet(mars,    4, rocky, 2).
planet(jupiter, 5, gas_giant, 95).
planet(saturn,  6, gas_giant, 146).
planet(uranus,  7, ice_giant, 28).
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gas_giant(P) :- planet(P, _, gas_giant, _).
has_moons(P) :- planet(P, _, _, M), M > 0.
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rocky_planets(L) :- findall(P, planet(P, _, rocky, _), L).
```

```
?- rocky_planets(L).
```

L = [mercury, venus, earth, mars]

```
% planet(Name, PositionFromSun, Type, NumberOfMoons)
planet(mercury, 1, rocky, 0).
planet(venus,   2, rocky, 0).
planet(earth,   3, rocky, 1).
planet(mars,    4, rocky, 2).
planet(jupiter, 5, gas_giant, 95).
planet(saturn,  6, gas_giant, 146).
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gas_giant(P) :- planet(P, _, gas_giant, _).
has_moons(P) :- planet(P, _, _, M), M > 0.
closer_than(P1, P2) :- planet(P1, A, _, _), planet(P2, B, _, _), A < B.
rocky_planets(L) :- findall(P, planet(P, _, rocky, _), L).
```

```
?- planet(saturn, Position, _, _).
```

Position = 6